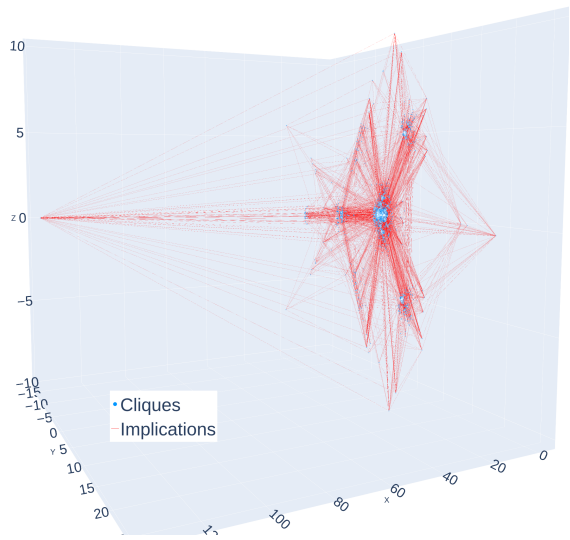

Figure 7: A global view of the latent space of equational theories and of the 1415 vertices and 4824 edges of the implication graph $(G, \rightsquigarrow)$ modulo reversibility. The existence of implication flows reveals an emerging similitude between the model-theoretic definition of the latent space and the proof-theoretic definition of the implication graph.

consequence: it enables us to derive from the original embedding of $(G, \rightsquigarrow)$ an embedding of the graph $(G, \rightsquigarrow)$ modulo $\sim$. The idea is to replace each reversible clique modulo $\sim$ by a single ball located at the center of mass of the associated cluster. The number of provably equivalent theories in the clique is indicated by the diameter of the ball. One obtains in that way a graph with 1415 vertices (blue balls of different diameters) and 4824 edges (red arrows), see Fig. 7.

We mention a number of remarkable emerging phenomena which appear when one observes the embedding of the graph $(G, \rightsquigarrow)$ modulo $\sim$.

8.1 Radial direction of implication flows

A striking observation on the picture of Fig. 7 is the existence of clearly oriented implication flows across the latent space. The mainstream direction goes along the X-axis towards equational theories with higher expectations. There is also a clear radial orientation along the Y and Z-axis towards theories with higher variance, and with different degrees of self-conjugacy.

The implication graph looks disorganised close to the source $\mathsf{Eqn}[1]$ which implies all the other theories. On the other hand, it looks very organised close to the sink $\mathsf{Eqn}[0]$ which is implied by every theory. One tentative explanation is that more equational theories $\mathsf{Eqn}[k]$ such as *idempotency* or *commutativity* can be deduced as consequences of a theory $\mathsf{Eqn}[j]$ closer to the source. These equational theories can be used in the reasoning to establish unexpected facts, and produce disorganised edges. On the other hand, when one gets closer to the sink, it becomes much more difficult to prove things, and this could explain why there are so few implication edges going against the mainstream radial orientation

8.2 Parallelism of implications

Another striking phenomenon is that many implication edges appear to be similar and parallel in the latent space. A closer inspection shows that very often, the parallel edges have similar source and target theories. An illustration is provided by the pair of parallel implications

$$\begin{array}{c}\mathsf{Eqn}[422] \Longrightarrow \mathsf{Eqn}[2851]\\ \mathsf{Eqn}[427] \Longrightarrow \mathsf{Eqn}[2861]\end{array}$$

where $\mathsf{Eqn}[422]$ and $\mathsf{Eqn}[427]$ are very similar:

$$\begin{array}{lll}\forall x.y.z. & x = x \diamond (x \diamond (y \diamond (z \diamond y)))\\ \forall x.y.z. & x = x \diamond (y \diamond (x \diamond (x \diamond z)))\end{array}$$

and for $\mathsf{Eqn}[2851]$ and $\mathsf{Eqn}[2861]$:

$$\begin{array}{lll}\forall x.y. & x = ((x \diamond (x \diamond y)) \diamond x) \diamond x\\ \forall x.y. & x = ((x \diamond (y \diamond x)) \diamond x) \diamond x\end{array}$$

We conjecture that this parallelism phenomenon is related to the Herbrand theorem in formal logic, see (Miller, 1987) and the discussion in Appendix. Note in particular that the witnesses (or expansion trees) provided by the Herbrand theory generalise the notion of *simple rewrite* elaborated and collected in Section 21 of the ET project blueprint. In that prospect, it would be worth collecting a large number of such parallel edges in the latent space and exploring how the underlying proofs are related, typically by inspecting their expansion trees, to see whether our conjecture holds. We leave that for future work.

8.3 Contrarian edges and hard implications

After observing the mainstream orientation of implication flows, we looked for atomic edges with a minority or contrarian orientation. To that purpose, we searched and collected a number of *longest paths* in the implication graph $(G, \rightsquigarrow)$ modulo $\sim$. Among these paths, we observed some unexpected behaviours like *swirls around the cliques or clusters* of provably equivalent theories. A possible explanation is that these clusters of provably equivalent theories are *active proof-theoretic areas* of the latent space, where many different equivalent